Aditya Nawal

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Education

National University of Singapore (NUS) **Jan 2025 – Jan 2029**
PhD, Department of Electrical and Computer Engineering *CGPA: 4.5/5.0*

Birla Institute of Technology and Science, Pilani **Aug 2018 – June 2022**
Bachelor of Engineering, Electronics and Electrical Engineering *CGPA: 8.39/10*

Research Publications

Relevance as a Vulnerability: How Web Retrieval Degrades Safety Alignment in LLM Agents [PDF]
Aditya Nawal, Manit Baser, Mohan Gurusamy
Findings of the Association for Computational Linguistics: EMNLP 2026 **Oct 2026**
Built AgentREVEAL to diagnose retrieval-induced safety degradation in LLM agents; identified relevance-triggered failure modes including the Safe Source Paradox, and introduced HarmURLBench with 1,405 real-world URLs and 320 harmful behaviors.

Revolutionizing visuals: The role of generative AI in modern image generation [PDF]
Gaurang Bansal, **Aditya Nawal**, Vinay Chamola, Norbert Herencsar
ACM Transactions on Multimedia Computing, Communications and Applications **Nov 2024**
Surveyed modern generative image-generation systems and analyzed their role in multimedia content creation, including quality, diversity, interpretability, computational efficiency, and practical deployment considerations.

An IoMT-based incremental learning framework with a novel feature selection algorithm for intelligent diagnosis in smart healthcare [PDF]
Siva Sai, Kartikey Singh Bhandari, **Aditya Nawal**, Vinay Chamola, Biplab Sikdar
IEEE Transactions on Machine Learning in Communications and Networking **Mar 2024**
Developed an IoMT-based incremental learning framework for smart healthcare diagnosis, improving anomaly detection and classification accuracy while reducing runtime versus batch-learning baselines.

Ongoing Works

Fetch or Refuse: What Governs Retrieval Admission [arXiv]
Aditya Nawal, Manit Baser, Mohan Gurusamy
Preprint **arXiv**
Analysis of what governs whether tool-using language models admit external sources during retrieval.

SLIP-ing into Harm: Multilingual Safety Across Tool Admission and Generation in LLMs [arXiv]
Aditya Nawal, Manit Baser, Mohan Gurusamy
Preprint **arXiv**
Cross-lingual safety behaviour of tool-using language models across the admission and generation stages.

In Tokens We Trust: Exposing Superficial Editing and Unlearning with Adversarial Tokenization [arXiv]
Manit Baser, **Aditya Nawal**, Mohan Gurusamy
Preprint **arXiv**
Security evaluation of post-release knowledge modification in open-weight language models.

PlaceCarry: Evaluating Placeholder-Based Semantic Carryover in Multi-Stage LLM Jailbreaks
Xiaodong Shi, **Aditya Nawal**, Mohan Gurusamy
Under review **ACL ARR, Aug 2026**
Multi-turn safety evaluation of language models under indirect, context-dependent references.

Trust and Provenance in Multi-Agent Systems **In preparation**

How agreement between collaborating agents affects the safety and provenance of what they produce.

Adaptive Multi-Turn Adversarial Evaluation of Tool-Using Agents **In preparation**

Extending retrieval-driven safety analysis to adaptive, multi-turn interaction over live web sources.

Cost and Resource Integrity in LLM Agent Serving **In preparation**

How untrusted context shapes the serving-time cost of agent workloads, and the policies that bound it.

Experience

National University of Singapore (NUS) Jan 2025 – Present

PhD Researcher, Department of Electrical and Computer Engineering

- Researching safety vulnerabilities in **retrieval-augmented LLM agents** and developing robust AI alignment methodologies.
- Conducted **mechanistic interpretability** on LLM agents to inspect internal representations and belief shifts during adversarial persuasion.

Microsoft July 2022 – December 2024

Software Development Engineer

- Engineered **Azure OpenAI-assisted reasoning workflows** to automate network failure diagnosis, enhancing consistency and resolution speed.
- Developed production tooling using **C#** and **React** to integrate LLM capabilities into internal cloud infrastructure services.
- Automated deployment workflows and onboarding pipelines for network operations utilizing infrastructure-as-code automation.

Microsoft Jan 2022 – June 2022

Software Development Intern

- Designed and implemented a centralized service marketplace platform to streamline internal tool discovery and cloud service usage.
- Consolidated fragmented software catalog services to improve developer experience and coordination across provider teams.

Akamai Technologies May 2021 – July 2021

Software Development Intern

- Developed automated security tooling to test DNS query routing and validate packet handling integrity across security boundaries.
- Configured simulated environments utilizing the **TOR network and Onion routing protocols** for anonymity assessment.
- Engineered network packet captures (pcaps) replicating threat scenarios in the **MITRE ATT&CK framework** to test defense signatures.

Projects

Copilot Skill For Network Troubleshooting and Remediation **Azure OpenAI**

- Developed an intelligent **Copilot Skill** using **C#** and **Semantic Kernel** for autonomous network fault diagnosis via natural language.
- Built integration pipelines using Azure Functions to transform diagnostic telemetry into structured, actionable troubleshooting plans.

Technical Skills

Core Languages & Frameworks: Python, C++, PyTorch, JAX, TensorFlow, Git, LaTeX, C#, React, SQL
Research & Expertise: LLM Alignment (RLHF, DPO), AI Safety & Red Teaming, Adversarial Machine Learning, Mechanistic Interpretability, Multi-Agent Systems, RAG Security, Multilingual NLP
Agentic & LLM Frameworks: AutoGen, Semantic Kernel, Swarm, Magentic-One, LangChain, Hugging Face, OpenAI API

Honors and Service

Leadership and Service: Led NSS BITS Pilani fundraising initiatives for underprivileged education and coordinated Microsoft IDC giving programs supporting child welfare, girl-child education, and disability-focused organizations.